

A Digital Twin-Based Framework for Precision Tomato Cultivation:

Integrating XGBoost-Based Prediction with Model Predictive Control

Dong Kyu Lee ^a, Bo Kyoung Kang ^b, Ye Jin Kim ^c, Tae Gyeom An ^b, Young Tae Chae ^{b,*}

^a Quantum AI Task, CTO division, LG Electronics. Co., Ltd., 19, Yangjae-daero 11-gil, Seocho-gu, Seoul, Republic of Korea

^b Department of Architectural Engineering, Gachon University, Gyeonggi-do 13120, Republic of Korea

^c Department of Biosystems Engineering, Seoul National University, Seoul 08826, Republic of Korea

* Corresponding authors.

E-mail addresses: ychoe@gachon.ac.kr (Y.T. Chae).

Abstract

This study presents a digital twin-based framework for precision tomato cultivation, demonstrating the integration of XGBoost-based growth prediction with model predictive control. The framework integrates three core components: a virtual control module (VCM) for real-time monitoring and equipment actuation; an environment twin (ET) validated with sensor data; and an XGBoost-based tomato growth prediction model with model predictive control (MPC).

The VCM, developed on a robust commercial game engine, ensured seamless synchronization between physical and virtual components. The ET, developed in EnergyPlus, accurately replicated greenhouse dynamics with minimal errors, well within standard thresholds. Correlation analysis was used to identify key environmental drivers of tomato crop responses, which informed the configuration of the growth prediction model. The XGBoost model was selected for its superior accuracy and robustness.

On-farm verification showed that the MPC significantly improved microclimate regulation, reducing temperature and humidity deviations by 70–80% compared to heuristic control. This stabilization led to substantial increases in productivity; total yield rose by 35.1%, while premium-grade crop production more than doubled. These findings confirm that this integrated framework provides a robust and scalable solution for precision tomato cultivation, enabling simultaneous improvements in yield, fruit quality, and resource efficiency.

Keywords : Digital twin, Precision Agriculture, Machine learning, Model predictive control

1. Introduction

Agriculture has served as a core foundation of human civilization, playing a critical role in food production and resource management. However, the sector is currently confronted with unprecedented challenges, including rapid population growth, resource depletion, and climate change. Climate change has led to soil fertility degradation, salinization, and deterioration of irrigation water quality, thereby increasing pest and disease incidence and causing significant yield losses (Vijai et al., 2023). Concurrently, population growth has accelerated the depletion of arable land and freshwater resources, requiring global crop production to more than double by 2050 to feed an estimated 9.7 billion people (Sahu and Liu, 2023). Agriculture thus faces the dual imperative of ensuring food security while maintaining environmental sustainability (Rushchitskaya et al., 2024).

To address these challenges, smart farming technologies incorporating the Internet of Things (IoT), sensors, artificial intelligence (AI), robotics, automation, and data analytics are being actively developed (Sharma et al., 2024; Bissadu et al., 2025). These technologies enable continuous monitoring of environmental and crop parameters, data-driven predictive analytics, and optimization-based decision-making for precision agriculture (Assimakopoulos et al., 2025; Nimma et al., 2025). IoT-based monitoring and data acquisition systems are already deployed in many smart farms (Devarajan, 2025; Hosny et al., 2025). Many current systems, however, remain focused on real-time data collection and basic rule-based operations, with only limited incorporation of predictive analytics and optimization-driven approaches (Yassine et al., 2022). Achieving both food security and environmental sustainability requires a paradigm shift from basic monitoring toward advanced control strategies that integrate predictive analytics and optimization (Rizan et al., 2024; Miller et al., 2025; Hamidane et al., 2024).

Existing research in smart farming can be broadly categorized into three technological domains.

The first domain focuses on sensor- and IoT-based monitoring and data acquisition, primarily aimed at collecting and visualizing environmental (temperature, humidity, CO₂, and light), soil (moisture and electrical conductivity), and crop growth (image-based) data (Foutia et al., 2024; Srisawat et al., 2025; Guo et al., 2025; Huang et al., 2025; Pal et al., 2023; Singh et al., 2023; Schlagmann et al., 2023; Juleang et al., 2024). For example, a cloud-based monitoring and control platform for small-scale farms was implemented, enabling users to observe environmental data and manually control actuators via a web interface (Foutia et al., 2024). An IoT-enabled monitoring system was also developed to evaluate agricultural environmental conditions, demonstrating its effectiveness in comparing natural rubber latex with conventional mulching materials for improved farm management (Srisawat et al., 2025). In addition, a lightweight deep learning model was developed for high-precision leaf detection and counting, providing an efficient approach for image-based crop monitoring in smart farming systems (Guo et al., 2025). Additional work has advanced sensor capabilities and network efficiency, including bio-impedance sensors for fruit physiological status evaluation (Huang et al., 2025), data-reduction algorithms to extend network lifetime (Pal et al., 2023; Singh et al., 2023), miniaturized leaf wetness sensors for microclimate monitoring (Schlagmann et al., 2023), and multi-protocol integration (LoRa and 4G/Wi-Fi) for high-speed data transmission (Juleang et al., 2024). Despite these advances, early sensor-based systems largely provided passive monitoring without predictive analytics or proactive environmental control.

The second category involves automated environmental control and decision-support systems that integrate sensor data with actuators to regulate irrigation, ventilation, and lighting (Lanje et al., 2022; Salimath et al., 2025; Rao et al., 2023; Manikandan et al., 2022).

Many employed rule-based logic or simple algorithms, occasionally incorporating expert knowledge. For instance, combining government datasets with in-field sensor measurements improved data accuracy and enabled automated control when environmental thresholds were exceeded (Lanje et al., 2022). IoT-based soil moisture sensing with automated fertigation improved water and nutrient efficiency in banana cultivation and increased profitability (Salimath et al., 2025). An automated irrigation system connected to an Android application maintained optimal soil moisture via Wi-Fi monitoring (Rao et al., 2023). Intelligent control strategies, including fuzzy logic and proportional–integral–derivative (PID) controllers, were introduced for real-time management (Manikandan et al., 2022). While these approaches improved operational efficiency, their reliance on static thresholds and single-variable optimization limited responsiveness to complex environmental interactions and crop physiological dynamics.

The third and most recent category focuses on data-driven prediction and intelligent analytics, employing empirical models and machine learning/AI to forecast crop yield (Sari et al., 2019), disease outbreaks, and optimal growth conditions (Choi and Lee, 2025; Li et al., 2024; Priya et al., 2024; Elashmawy and Uysal, 2023; Yonbawi et al., 2023; Fan et al., 2015). For example, Non-linear models such as logistic and Gompertz functions were used to describe tomato production and identify inflection points as indicators of yield potential (Sari et al., 2019). CNN-predicted leaf area was integrated into greenhouse evapotranspiration models, ensuring robust accuracy and enabling uncertainty assessment for precision irrigation (Choi and Lee, 2025). In addition, hyperspectral imaging with a CNN-GRU model accurately estimated shoot and root biomass of tea seedlings, confirming the superiority of deep learning over traditional methods (Li et al., 2024). An XGBoost-based system achieved an R^2 of 0.92 for crop yield prediction through feature prioritization and hyperparameter optimization (Priya et al., 2024). Another study applied machine

learning to link soil properties with crop quality through multivariate correlation analysis (Elashmawy and Uysal, 2023). A metaheuristic-tuned classification algorithm improved irrigation demand prediction accuracy by categorizing water requirements into five levels (Yonbawi et al., 2023). In addition, a knowledge-and-data-driven modeling approach integrated the GreenLab model with a radial basis function network, achieving higher accuracy in tomato growth prediction than single-model approaches (Fan et al., 2015). Within this context, digital twin (DT) technology has emerged as a promising approach, enabling synchronized virtual representations of physical farms. However, most implementations have addressed only partial DT functionalities, focusing on monitoring or single-variable optimization without achieving fully integrated, real-time environmental control.

DT applications have gained particular traction in greenhouses and indoor farming systems, including plant factories and aquaponics. Research in this domain can be divided into three substreams. First, DT design studies for horticulture emphasize the integration of sensors, IoT, virtual reality (VR), augmented reality (AR), and cloud computing (Miller et al., 2025). For example, a DT prototype combining Sensing Change soil probes with the SWAMP IoT water management platform was developed to optimize irrigation (Alves et al., 2023). Second, DT-based plant growth studies monitor and predict growth stages, such as a smart aeroponic potato cultivation system integrating farm sensors, a cloud database, and visualization modules (Asfarian and Wulandari, 2023), aquaponics simulations predicting fish growth (Ghandar et al., 2021), and greenhouse production flow tracking from cutting arrival to shipment (Howard et al., 2022). A DT reference architecture was also proposed and validated using greenhouse crop data (Spyrou et al., 2024). Third, DT-based environmental control and energy optimization approaches include geometric ray-tracing and genetic optimization for solar greenhouse design (Isied et al., 2022), dynamic plant

factory control using digital replicas (Zhang et al., 2023), thermal prediction with support vector regression (SVR) (Hull et al., 2023), and real-time power consumption visualization using DT-based color-coded mapping (Liu et al., 2023).

Despite these contributions, a significant research gap remains in developing a holistic, real-time synchronized DT framework that not only monitors and predicts but also proactively and precisely controls complex agricultural environments to optimize crop-specific productivity and resource-use efficiency. This study addresses this gap by proposing, developing, and validating a comprehensive DT-based system for precise environmental control in tomato smart farms. By replicating the greenhouse structure and equipment within a fully functional DT that continuously integrates real-time sensor data, the system enables accurate prediction of key environmental variables and implements efficient control strategies to dynamically optimize growing conditions. Through experimental validation, this study demonstrates the benefits of such an integrated approach, including improved tomato yield and enhanced resource efficiency.

2. Method

2.1 Study site and crop

The study was conducted in two glass-enclosed greenhouses situated in Pyeongchang, Gangwon Province, Republic of Korea. The plant material consisted of tomato (*Solanum lycopersicum* 'Dairos'), which was cultivated over a 12-week experimental period.

A sensor network was deployed to continuously monitor environmental conditions. Internal sensors recorded air temperature, relative humidity (RH), and CO₂ concentration within the greenhouses, while external sensors measured ambient air temperature, RH, and solar radiation.

2.2 Digital twin-based greenhouse control system schema

A Digital Twin (DT) framework was developed to enable precise environmental control and predictive optimization of crop growth in greenhouse production systems. The framework consists of three core components: a Virtual-Control Module (VCM), an Environment Twin (ET), and a Growth Predictor (GP). As illustrated in Fig. 1, these components, while functionally distinct, operate synergistically and interdependently to form a cohesive digital twin-based greenhouse control system.

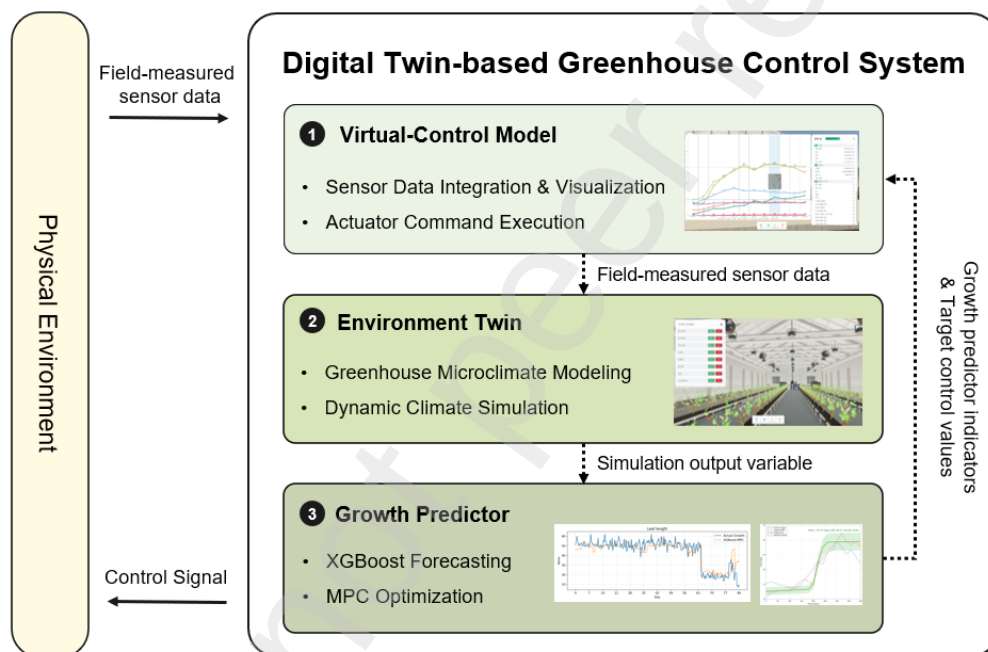


Fig. 1. Overview of the DT-based greenhouse control system.

The VCM, implemented on a real-time 3D development platform, provides data visualization, interactive simulation, and execution of control actions derived from predictive analytics. The ET serves as the core physics-based simulation engine for modeling thermal performance, airflow, lighting, and HVAC system dynamics. The GP was established through comparative evaluation of machine learning and deep learning models for forecasting agronomic responses to environmental variables, thereby enabling data-

driven optimization of cultivation parameters.

The operational workflow of the DT framework follows a continuous closed-loop process. Real-time sensor data from the physical greenhouse are acquired by the VCM and transferred to the ET for simulation. The resulting environmental outputs are processed by the GP, which forecasts crop growth and determines optimal setpoints. These setpoints are returned to the VCM, which translates them into actuator commands and transmits them to the physical control hardware. This iterative cycle maintains synchronization between the digital and physical systems, supporting model-based decision-making for greenhouse management.

2.3. Virtual-control module

2.3.1. Virtual greenhouse modeling and visualization

The VCM was developed on Unity, a real-time 3D platform (Unity Technologies, 2023), to support interactive visualization and simulation of the greenhouse environment. A virtual model was constructed to accurately model the greenhouse's geometric structure, spatial configuration and key operational components, establishing a direct one-to-one correspondence between the physical facility and its digital representation.

Within this virtual environment, dynamic conditions such as temperature distribution, solar radiation patterns and airflow dynamics are simulated and rendered in real time. This functionality supports both qualitative visual inspection and quantitative analysis of spatio-temporal environmental variations. The fidelity of this replication is illustrated in Fig. 2, which presents side-by-side images of the physical greenhouse used in this study and its digital twin counterpart constructed to replicate it.

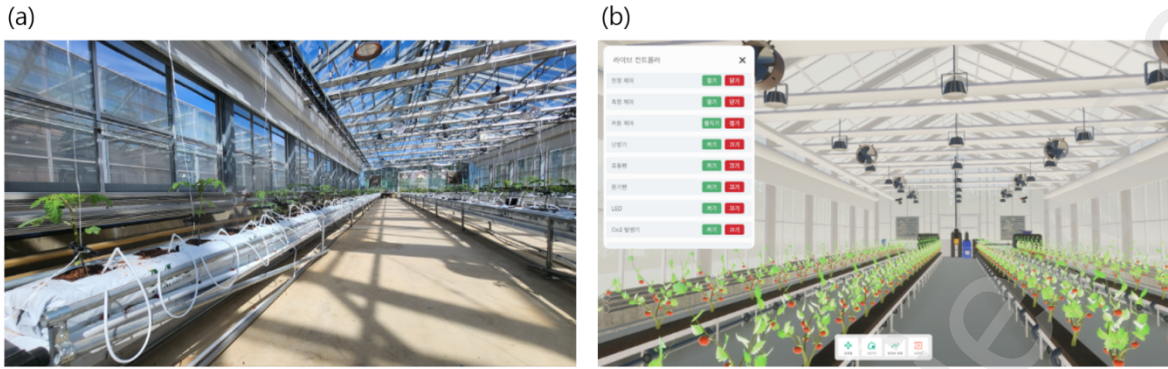


Fig. 2. Internal view of the greenhouse: (a) actual greenhouse interior; (b) virtual representation of the interior within the 3D simulation environment.

2.3.2 Realtime synchronization and control

The primary function of the VCM is to maintain synchronization between the physical greenhouse and its digital twin, and to execute control commands securely. To achieve this, the module continuously acquires sensor data (including temperature, humidity, CO₂, solar radiation, and actuator statuses) from the physical asset via secure HTTPS communication pipelines. This data stream ensures the digital twin is perpetually updated to reflect real-world conditions. In the reverse data path, optimal setpoints from the GP are translated into low-level commands for actuators (e.g., heating units, humidity controls, CO₂ controls). These commands are then transmitted to the greenhouse control hardware using secure RESTful API calls, ensuring standardized and reliable implementation of the control strategy.

This synchronized data is presented to the operator through a real-time 3D visualization interface. The interface provides an immersive virtual environment that models the greenhouse's physical layout and equipment. It supports interactive features, such as a ray-casting mechanism for equipment selection and inspection, and dynamic skybox rendering to visualize external lighting conditions. This functionality allows for enhanced situational awareness, direct monitoring of environmental variables and equipment states,

and intuitive interaction with the digital twin-based greenhouse control system.

2.4 Environment twin

2.4.1 Simulation engine implementation

The ET was implemented using EnergyPlus, a dynamic building energy simulation engine widely applied for modeling complex physical interactions with high temporal resolution (U.S. Department of Energy, 2024). The engine performs time-step-based simulations, which, unlike steady-state analysis, can capture transient environmental variations. This capability is essential for the rigorous evaluation of dynamic control strategies, a requirement also emphasized in recent greenhouse studies employing transient models for HVAC system optimization (Çam et al., 2024).

Its modular architecture facilitated the detailed specification of the greenhouse model, including structural envelope components, glazing systems, and HVAC equipment configurations in Fig. 3. This approach enabled a replication of the greenhouse's thermal performance, airflow patterns, and lighting conditions, providing a robust foundation for simulation-based control.

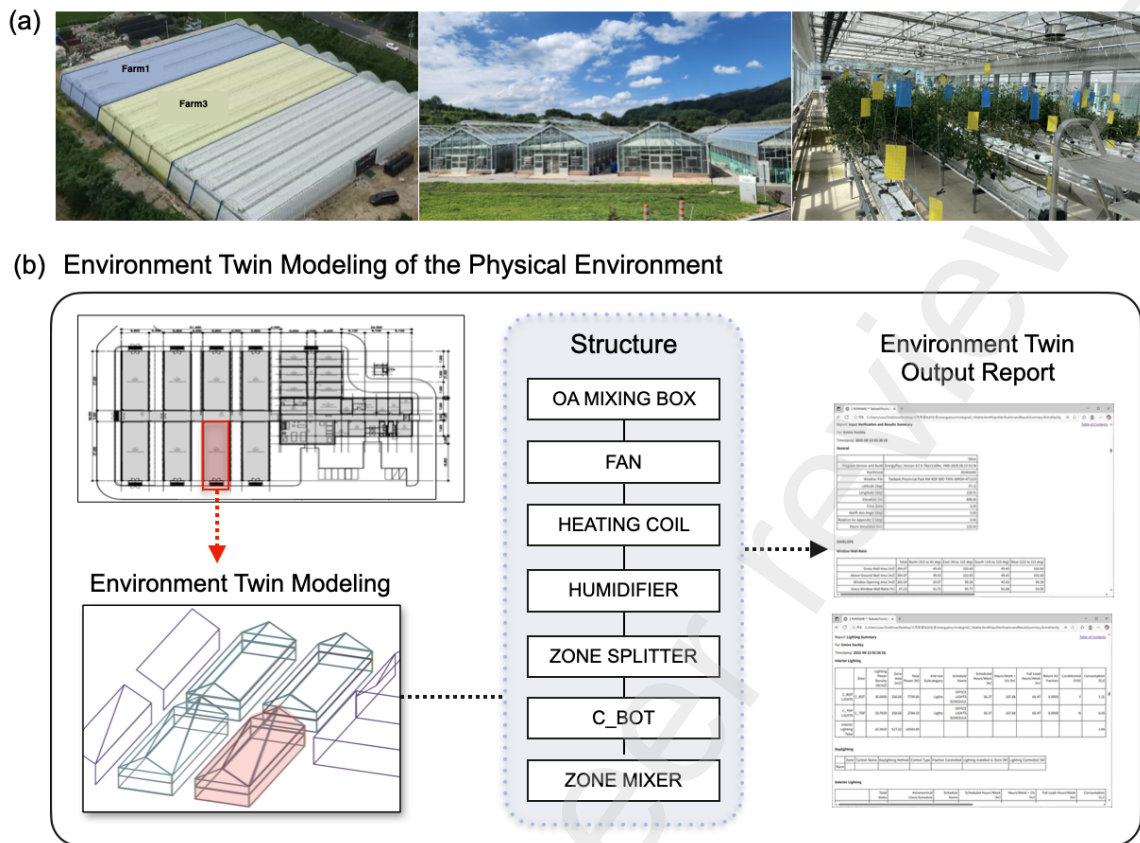


Fig. 3. (a) Physical greenhouse photograph and layout; (b) corresponding ET model.

2.4.2 Model parameterization and operational configuration

The ET model was parameterized using a set of input data files (IDFs) that defined its key structural and thermal properties. These parameters, including building envelope materials, internal zoning, and glazing characteristics, are detailed in Table 1. To align the simulation with real-world conditions, the model was driven by a combination of standard meteorological data in EPW format and field-measured sensor data from the physical greenhouse. Furthermore, key operational sub-systems were modeled to replicate the greenhouse's climate control dynamics, including a ventilation system (roof/side vents and mechanical fans), an electric unit heater, and a dedicated CO₂ injection module.

Table 1

207 Structural geometry, envelope materials, and thermal properties of the virtual greenhouse.

	Field	Value
Structural Geometry	Floor Area	517.32 m ²
	Volume	1334.81 m ³
	Gross Window-Wall Ratio	87.23 %
	Skylight-Roof Ratio	95.64 %
Envelope Materials	Glazing System	Double glazing (clear)
	Layer stack	Clear glass(3mm) / Air gap(13mm) / Clear glass(3mm)
Thermal Propertys	U-Factor	2.72 W/ m ² -K
	SHGC	0.764

208

209 The primary outputs of the simulation—time-series data for indoor temperature,
 210 relative humidity, and CO₂ concentration—were transmitted to the GP for agronomic
 211 forecasting. These outputs ensured consistency between the simulated environmental
 212 conditions and the input variables required for crop growth prediction.

213 2.5. Growth predictor

214 2.5.1. Dataset and preprocessing

215 For model training, we utilized a publicly available open-source dataset (RDA Smart
 216 Farm), which encompassed 58,624 environmental measurements and 353 corresponding
 217 crop growth observations. The test dataset (SNU Pyeongchang Campus), in contrast, was

constructed from a combination of simulation outputs generated by an Environment Twin and proprietary real-world growth data acquired during the period of 2023–2024.

The input features comprised six key environmental parameters: indoor temperature ($^{\circ}\text{C}$), indoor humidity (%), indoor CO_2 concentration (ppm), outdoor temperature ($^{\circ}\text{C}$), outdoor relative humidity (%), and solar radiation (W/m^2). The output variables represented eight critical agronomic indicators of crop growth and yield: growth length (cm), leaf length (cm), leaf breadth (cm), number of leaves (-), stem thickness (mm), flower height (cm), number of flower clusters (-), and number of fruit clusters (-).

A systematic data preprocessing pipeline was implemented to ensure the quality and integrity of the dataset. Initially, missing numerical values were imputed using feature-wise mean substitution. Subsequently, a forward-fill strategy was applied to preserve temporal continuity in the data. Outlier management was performed using the interquartile range (IQR) method; specifically, data points falling below $Q1 - 1.5 \times \text{IQR}$ or exceeding $Q3 + 1.5 \times \text{IQR}$ were capped at these respective boundaries, where $Q1$ and $Q3$ denote the first and third quartiles. For feature scaling, input variables were standardized via Z-score transformation, while output variables were individually scaled using a robust scaler to minimize the influence of extreme values. Critically, all parameters for these preprocessing steps (e.g., mean, standard deviation, quartiles) were derived solely from the training dataset to prevent any information leakage into the test set.

2.5.2 Model selection of growth prediction

To select the optimal predictive model, a comparative evaluation was performed on seven candidate algorithms. This cohort included five tree-based ensemble methods—Random Forest, XGBoost, LightGBM, CatBoost, and Extra Trees—and two deep learning architectures renowned for modeling sequential data: Long Short-Term Memory (LSTM)

and Transformer. This selection was predicated on their established efficacy in managing high-dimensional, noisy sensor data and capturing complex temporal dependencies.

Specifically, Random Forest and Extra Trees were incorporated as robust ensemble methods capable of capturing non-linear environment–growth interactions while remaining resilient to sensor noise (Breiman, 2001; Liaw and Wiener, 2002; Geurts et al., 2006). XGBoost and LightGBM were included due to their strong performance on large-scale structured sensor datasets, efficient handling of missing values, and scalability for real-time control and IoT-based applications (Chen and Guestrin, 2016; Ke et al., 2017). CatBoost was selected for its advanced treatment of categorical features, allowing effective integration of variables such as growth stages and control settings into predictive modeling (Prokhorenkova et al., 2018). Beyond tree-based ensembles, LSTM was adopted for its ability to model long-term temporal dependencies through recurrent gating mechanisms (Hochreiter and Schmidhuber, 1997), while the Transformer was chosen for its self-attention mechanism, which enables learning of complex spatiotemporal patterns across multiple scales without recurrence (Vaswani et al., 2017).

Taken together, these models span both classical ensemble algorithms and state-of-the-art deep learning approaches, thereby ensuring that the diverse characteristics of greenhouse data—including non-linearity, missing values, categorical variables, and temporal dependencies—are comprehensively addressed in the comparative evaluation.

2.5.3 Correlation performance of prediction models

Correlation analyses were conducted between environmental variables and crop growth indicators, including pairwise Pearson correlations, lagged cross-correlations up to ± 48 hours, and partial correlations controlling for confounders (e.g., solar radiation). These results guided feature screening, prioritized input–output associations for control synthesis,

and contextualized the interpretation of model behavior.

The candidate models introduced in Section 2.5.2, trained on the dataset described in Section 2.5.1, were then evaluated using ET-generated environmental trajectories as inputs. Model forecasts were systematically evaluated against crop growth measurements collected during 2023–2024, with a three-week subset designated for validation. The model demonstrating the closest agreement with real-world observations was selected as the final predictive engine for MPC integration. The predictive mapping is expressed as:

$$\hat{y}_\tau = f_\theta(u_\tau, d_\tau), \quad (1)$$

where $\hat{y}_\tau$ denotes the vector of predicted agronomic outputs at timestep τ ; f_θ represents the selected predictive model parameterized by θ ; u_τ is the vector of controllable inputs (e.g., indoor temperature, humidity, CO₂ setpoints); and d_τ is the vector of uncontrollable external disturbances (e.g., solar radiation, outdoor weather conditions).

2.5.4 MPC based control design

The selected predictive engine was embedded into a Model Predictive Control (MPC) framework to compute optimal control trajectories over a finite horizon. The optimization problem is formulated as:

$$\min_u \sum_{\tau=t}^{t+H-1} |y_\tau^{target} - \hat{y}_\tau(u)|_Q^2 + |\Delta u_\tau|_R^2 \quad (2)$$

where $\hat{y}_\tau(u)$ denotes the predicted agronomic outputs at timestep τ , y_τ^{target} the corresponding targets, $\Delta u_\tau = u_\tau - u_{\tau-1}$ the incremental control adjustment, and Q and R diagonal positive semi-definite weighting matrices for tracking accuracy and control effort, respectively. The first term enforces adherence to agronomic objectives, while the second prevents abrupt actuator changes to ensure stability and energy efficiency.

A hybrid weighting strategy was adopted for Q . Baseline weights derived from

sensitivity analysis of the training dataset capture long-term structural dependencies, whereas dynamic modulation factors estimated online reflect transient, context-specific variations. This integration enables the controller to remain robust to persistent system relationships while adaptively responding to real-time disturbances.

Operational feasibility was further ensured through the following policies: (i) actuator adjustments were limited to $\pm 15\%$ of the operational range per cycle, with all inputs bounded within empirically validated safe envelopes; (ii) phase-aware weighting activated distinct Q profiles for vegetative (elongation/leaf expansion), structural (stem thickening), and reproductive (flowering/fruit set) stages; and (iii) execution cadence was aligned with physiological response times, with sensors sampled every 5–10 minutes and control actions updated every 30–60 minutes. Correlation analyses and sensitivity weights were recalibrated weekly using a rolling window with a fixed hold-out set to detect distributional drift.

2.5.5 Online learning and adaptation

The framework incorporates online learning to adapt to evolving environmental and crop dynamics. Model parameters are continuously updated using real-time sensor data streamed from the VCM, and forecasts can be generated at configurable intervals ranging from 1 to 24 hours. Prediction robustness is strengthened through uncertainty estimation and confidence interval analysis based on variance in recent prediction errors.

Continuous performance monitoring is conducted through real-time error tracking and distributional drift detection. When predefined thresholds are exceeded, automated retraining is triggered to restore predictive reliability. All forecasts, observed growth data, and executed control actions are systematically logged to ensure traceability, interpretability, and post-hoc analysis of system behavior.

2.6 System architecture

The system architecture is a closed-loop framework composed of three core components: a VCM, an ET, and a GP. This structure facilitates a continuous, iterative process of simulation, prediction, and control. The operational cycle begins with the VCM acquiring real-time sensor data from the physical greenhouse, including parameters such as temperature, humidity and actuator statuses. This data is then transmitted to the ET—a physics-based digital replica of the production environment—which simulates future environmental conditions based on the current state and applied control inputs.

The simulated environmental variables serve as input for the GP. This component utilizes trained predictive models to forecast crop growth trajectories and, based on these forecasts, computes optimal environmental setpoints required to achieve predefined agronomic targets. These setpoints are subsequently returned to the VCM, which translates the high-level targets into low-level, device-specific actuator commands. The module determines the appropriate control signals for equipment such as heating units and ventilation systems, which are then securely transmitted to the greenhouse control hardware via a RESTful API.

Execution of these commands alters the greenhouse environment. The resulting changes are captured by the sensor network and fed back to the VCM, thus closing the feedback loop. This iterative process ensures continuous synchronization between the digital twin and the physical asset, allowing control actions to be perpetually informed by predictive analytics. By tightly integrating growth forecasting with operational control, the framework enables proactive and adaptive management of the digital twin-based greenhouse control system, forming the basis for the performance evaluation presented in the Results section.

2.7 Evaluation metrics

2.7.1 Calibration metrics for environment twin

The calibration accuracy of the ET was evaluated using the Coefficient of Variation of the Root Mean Square Error (CV(RMSE)), a widely recommended metric for building energy model validation (ASHRAE, 2014). CV(RMSE) is obtained by normalizing the root mean square error (RMSE) by the mean of the observed values, providing a relative measure of the deviation between simulated and measured data. Lower CV(RMSE) values indicate closer agreement between simulation and observation, reflecting higher model fidelity. The CV(RMSE) is calculated as:

$$CV(RMSE) = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}}{\bar{y}} \times 100 \quad (3)$$

where y_i is the measured value, $\hat{y}_i$ is the simulated value, $\bar{y}$ is the mean of measured values, and n is the number of data points.

2.7.2 Prediction metrics for growth predictor

The predictive performance of the GP was evaluated using two metrics: RMSE and Mean Absolute Error (MAE). RMSE is calculated as the square root of the Mean Squared Error (MSE), thereby expressing prediction errors in the same units as the target variable and providing an intuitive measure of typical error magnitude (Hodson, 2022). MAE is the mean of the absolute differences between predicted and observed values, offering a straightforward and interpretable accuracy measure that is less sensitive to outliers (Hodson, 2022; Willmott and Matsuura, 2005). The metrics are defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

where y_i is the measured value, $\hat{y}_i$ is the predicted value and n is the number of data

353 points.

354 2.7.3 Yield and quality assessment

355 Crop yield was quantified as the total harvested weight over the cultivation period,
356 with weekly yields recorded separately under MPC and heuristic control. Fruit quality was
357 classified at harvest according to commercial grading standards (3L, 2L, M-L, S, 2S, 3S),
358 based on fruit size and weight. Premium-grade yield was defined as the combined output of
359 3L and 2L classes. For each treatment, both the absolute premium yield and its proportion
360 relative to the total harvest were calculated.

361 3. Results

362 3.1 Implementation of VCM

363 The digital twin of the physical greenhouse was developed and validated to serve as
364 the core of the simulation-control framework. The digital twin's 3D environment accurately
365 represents the geometric structure, spatial configuration, and operational equipment of the
366 real-world facility, as shown in Fig. 4. To enhance realism, dynamic skybox rendering was
367 implemented to simulate diurnal cycles, providing visualizations of lighting conditions that
368 correspond to external climate data.

(a)



(a')



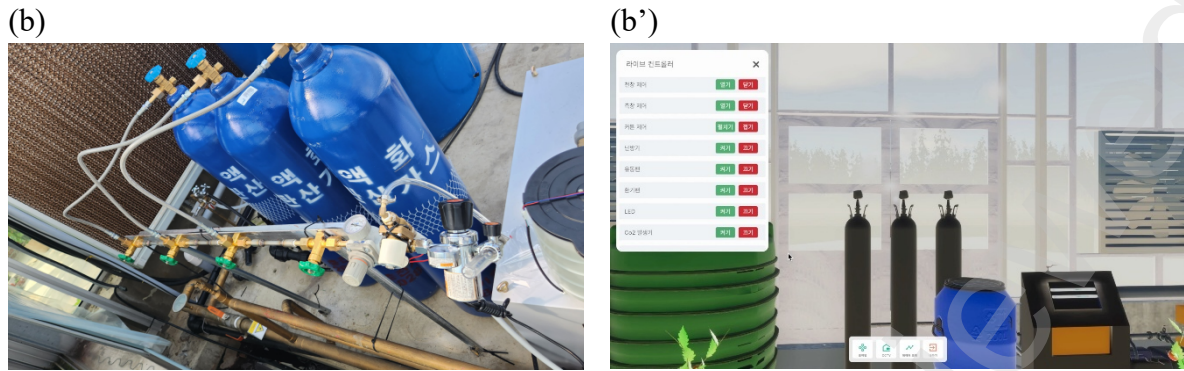
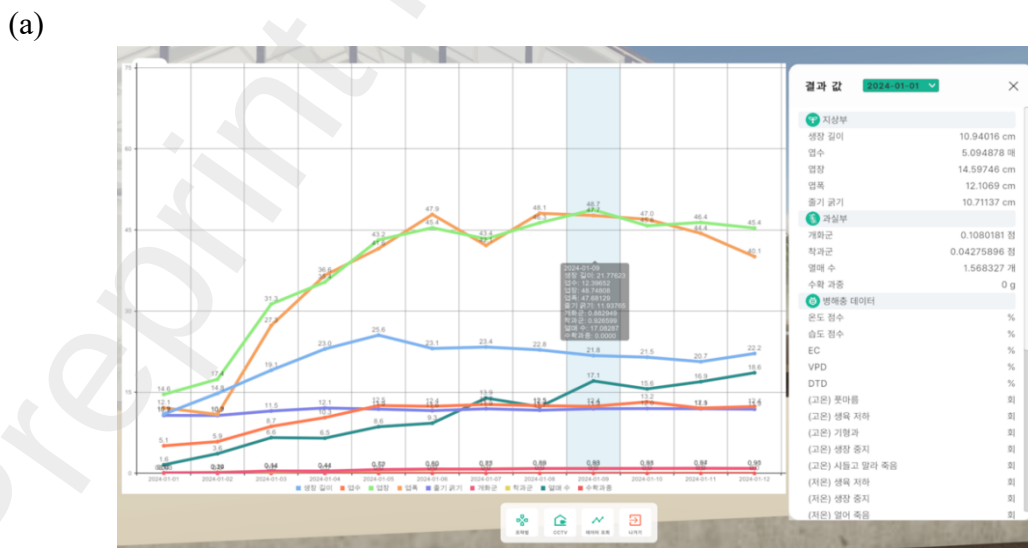


Fig. 4. Digital twin representation of greenhouse actuators: (a, a') heating unit and virtual model; (b, b') CO₂ injector and virtual model.

Key environmental control actuators, such as the CO₂ controller and heating unit, were functionally and visually integrated into the virtual environment. This integration enables real-time monitoring of actuator states, which are synchronized with their physical counterparts. The operational status of these devices, along with critical environmental variables (i.e., temperature, relative humidity, and CO₂ concentration), are visualized through a graphical user interface within the 3D environment, as shown in Fig. 5. A secure data pipeline ensures that these state variables are continuously updated, maintaining an operationally accurate and validated digital twin for predictive control and analysis.



(b)

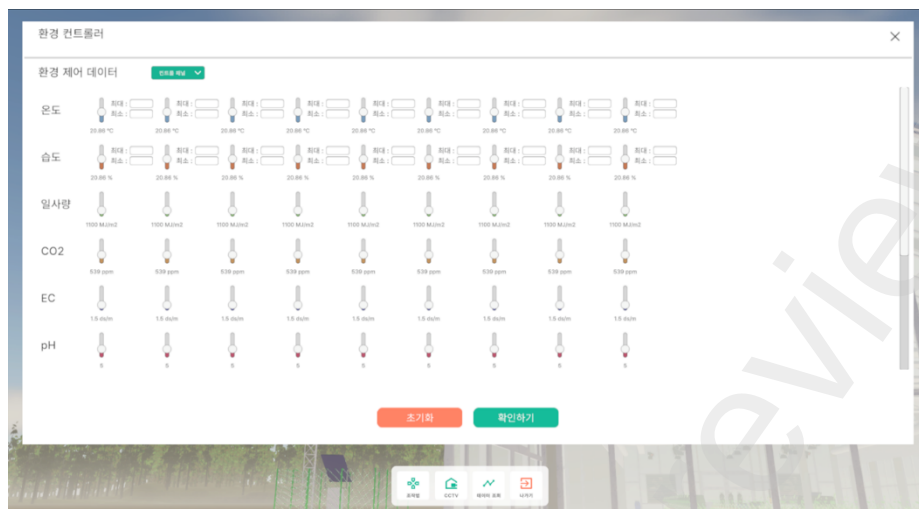


Fig. 5. Digital twin interface: (a) environmental variable graphs; (b) actuator status display.

3.2 Environment twin validation

The fidelity of the ET was quantitatively validated against sensor measurements obtained from the physical greenhouse across a five-day experimental period. The CV(RMSE) was calculated for key environmental variables, resulting in values of 0.089 for indoor air temperature, 0.116 for relative humidity, and 0.089 for CO₂ concentration. These values all fall well within the acceptable tolerance of 0.3 (ASHRAE, 2014).

As illustrated by the time-series comparison in Fig. 6, the model accurately captured the temporal dynamics of the internal greenhouse environment, demonstrating robustness across varying external weather and internal control conditions. These validation results confirm the ET's suitability as a simulator for use within the predictive control framework.

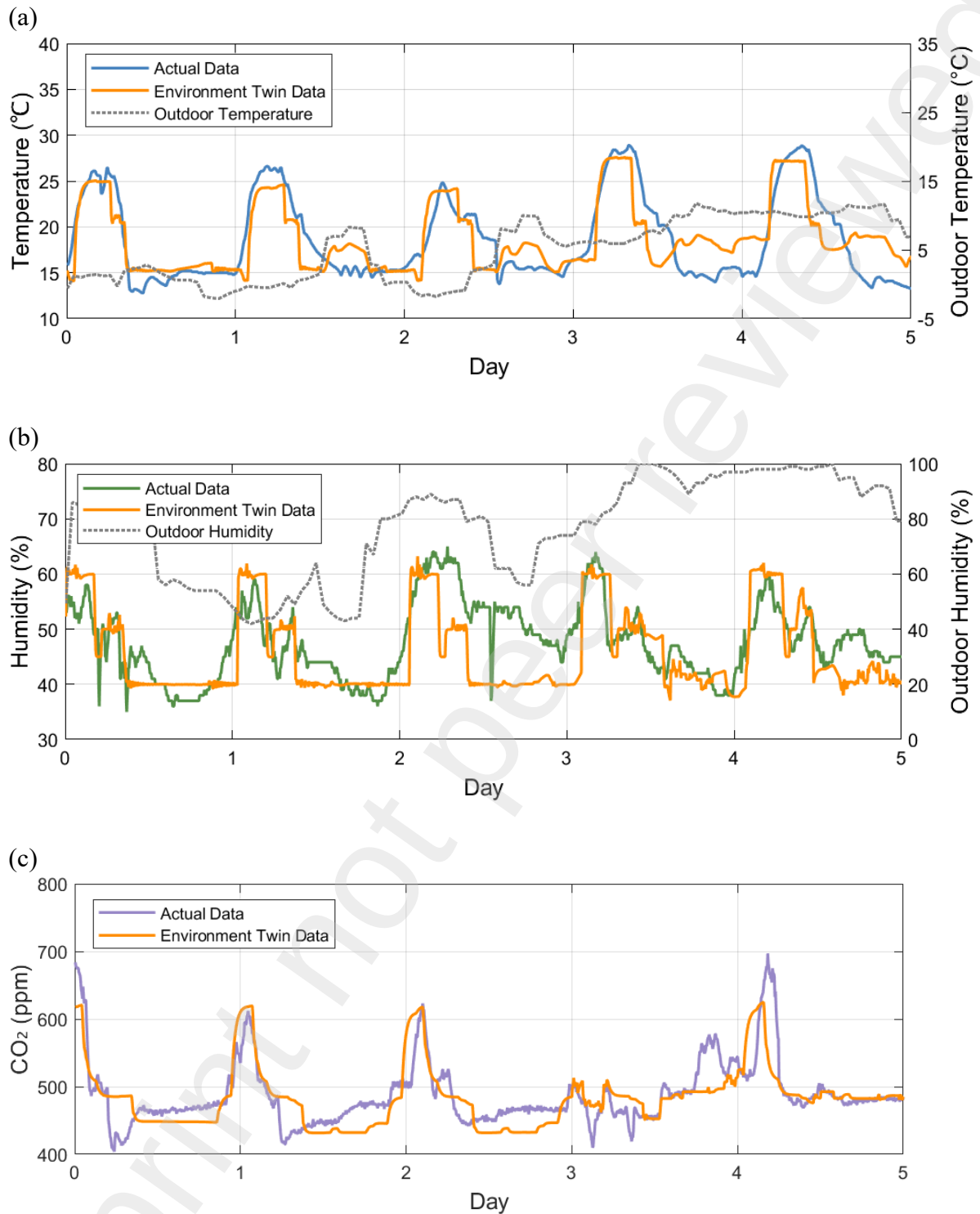


Fig. 6. Calibration of the ET model against measured data: (a) indoor air temperature; (b) relative humidity; (c) CO₂ concentration.

3.3. Growth predictor validation and integrating control

3.3.1 Correlation analysis

To elucidate the input–output relationships within the test dataset, a comprehensive correlation analysis was conducted. This included pairwise Pearson correlations, lagged cross-correlations (± 48 hours), and partial correlations to account for confounding factors such as solar radiation. The variables were classified into controllable indoor environmental parameters (e.g., temperature, Humidity, CO₂) and non-controllable exogenous variables.

As shown in Fig. 7, the most prominent statistical associations were observed with external weather conditions. Outdoor relative humidity exhibited strong positive correlations with leaf breadth ($r = 0.753$) and leaf length ($r = 0.692$), while also presenting the strongest negative correlation with flower height ($r = -0.819$). Solar radiation was positively correlated with growth length ($r = 0.678$). Among the controllable factors, indoor temperature had the greatest influence, showing a negative correlation with flower height ($r = -0.716$) and a strong positive correlation with stem thickness ($r = 0.626$). These results highlight the primary environmental drivers of crop development during the evaluation period.

The derived correlation structure was subsequently used to configure the control system by constraining admissible actuator–trait pairings and initializing sensitivity weights within the Model Predictive Control (MPC) framework. The MPC model then exploited these empirically observed relationships to predict optimal environmental setpoints for maximizing crop growth, thereby ensuring that the control strategy was directly informed by the specific dynamics of the cultivation environment.

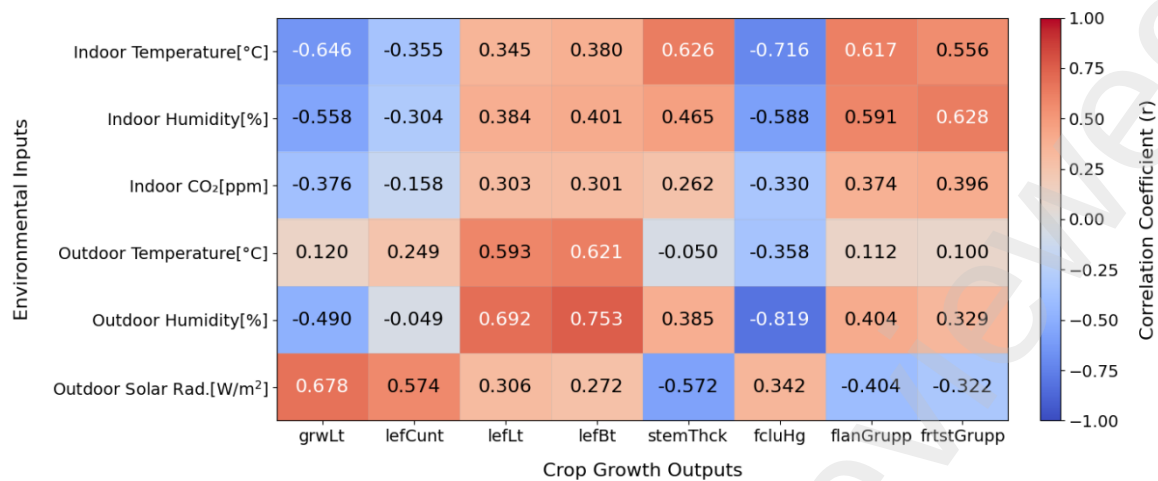


Fig. 7. Correlation between six environmental variables and eight crop growth indicators. Abbreviations: grwLt, growth length; lefCunt, leaf count; lefLt, leaf length; lefBt, leaf breadth; stemThck, stem thickness; fcluHg, flower height; flanGrupp, flower cluster count; frstGrupp, fruit cluster count.

3.3.2. Model comparative evaluation

The predictive performance of the evaluated models over the three-week validation period is summarized in Fig. 8, detailing the mean RMSE and mean MAE averaged across eight growth indicators. The XGBoost model achieved the highest accuracy, with a mean RMSE of 5.562 and a mean MAE of 3.766. This performance surpassed that of other tree-based models, including CatBoost (Mean RMSE = 5.703; Mean MAE = 3.926), LightGBM (5.865; 3.892), Random Forest (5.892; 3.882), and Extra Trees (6.038; 3.886). Specifically, XGBoost yielded a 2.47% lower mean RMSE than CatBoost and a 2.99% lower mean MAE than Random Forest. In comparison, the ensemble (8.644; 7.204) and deep learning models, including LSTM (14.420; 13.176) and Transformer (18.025; 15.460), produced markedly higher errors. These results confirm the superior suitability of tree-based boosting methods for this forecasting task. Accordingly, XGBoost was selected as the final predictive engine.

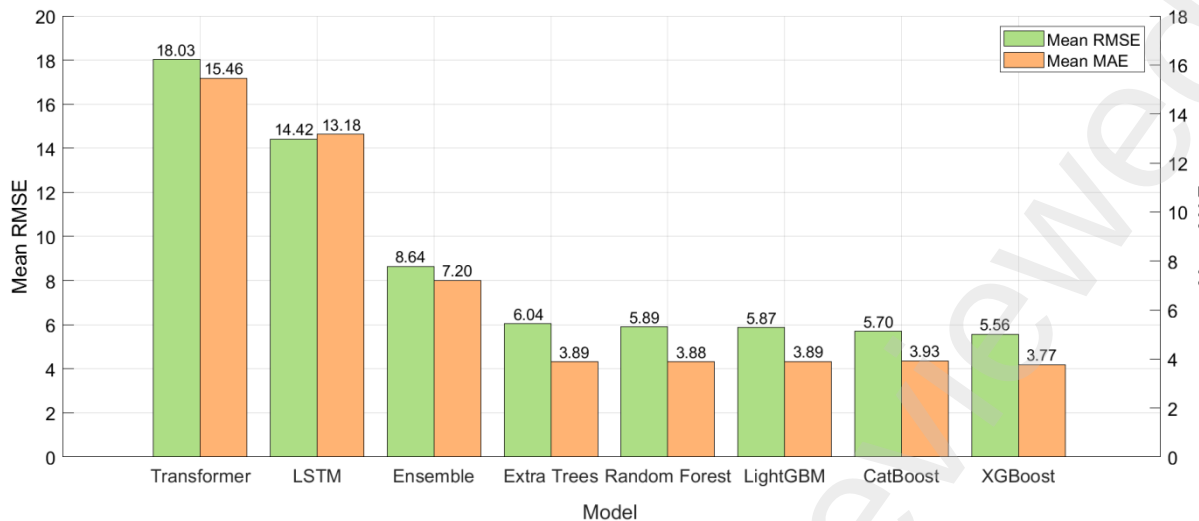


Fig. 8. Comparative mean RMSE(left) and MAE(right) of seven models across growth indicators.

3.3.3 Trait-level accuracy of the XGBoost-MPC

Trait-level predictive performance was evaluated over the three-week validation period, as summarized in Table 2 and illustrated in Fig. 9. Vegetative and structural traits demonstrated consistently high accuracy. Stem thickness (RMSE = 0.42 mm; MAE = 0.31 mm) exhibited the lowest error values, while leaf length (6.83 cm; 4.93 cm), leaf breadth (7.15 cm; 5.10 cm), and leaf count (2.39; 1.64) were reproduced with strong fidelity. Growth length (11.27 cm; 7.58 cm) showed comparatively larger deviations but remained within an acceptable range for cumulative growth indicators.

In contrast, reproductive traits exhibited lower predictive accuracy. Flower cluster height (15.59 cm; 9.98 cm) recorded the highest error values among all traits, reflecting the physiological complexity and long developmental horizons associated with reproductive processes. Although flowering cluster group (0.40; 0.27) and fruit set group (0.44; 0.32) achieved relatively small absolute errors, their predictive stability was affected by discrete transitions, such as the initiation of new clusters, which are inherently difficult to capture within short validation horizons.

449

450 **Table 2**

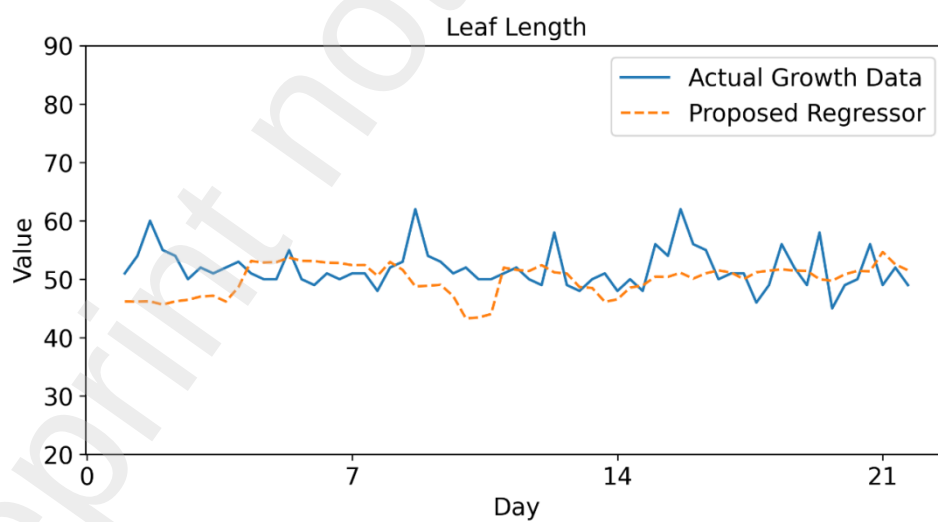
451 RMSE and MAE of structural and reproductive traits predicted by the XGBoost.

Metric	RMSE	MAE
Growth length [cm]	11.267	7.583
Leaf length [cm]	6.834	4.929
Leaf breadth [cm]	7.152	5.098
Leaf count [count]	2.388	1.637
Stem thickness [mm]	0.424	0.311
Flower cluster height [cm]	15.594	9.976
Flowering cluster group [count]	0.396	0.274
Fruit set group [count]	0.437	0.316

452

453

(a)



(b)

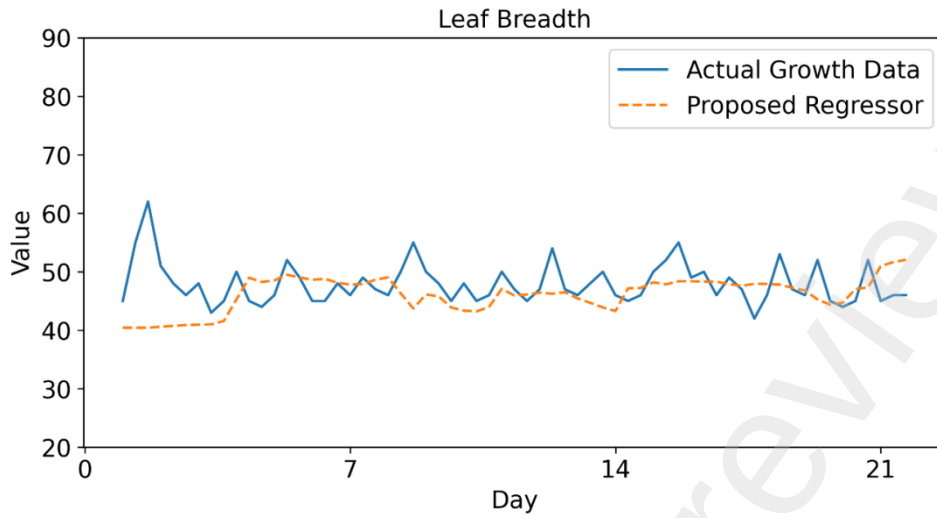


Fig. 9. Actual versus predicted time-series of (a) leaf length and (b) leaf breadth.

3.3.4 Control strategy comparison

A comparative analysis of actuator activity was conducted between the MPC and the heuristic control over a 16-day evaluation period, as shown in Fig. 10. A marked divergence was observed: the MPC executed a total of 1,809 control actions (585 for heating, 1,024 for CO₂, and 200 for humidity), while the heuristic control initiated only 508 actions (142 for heating, 174 for CO₂, and 192 for humidity).

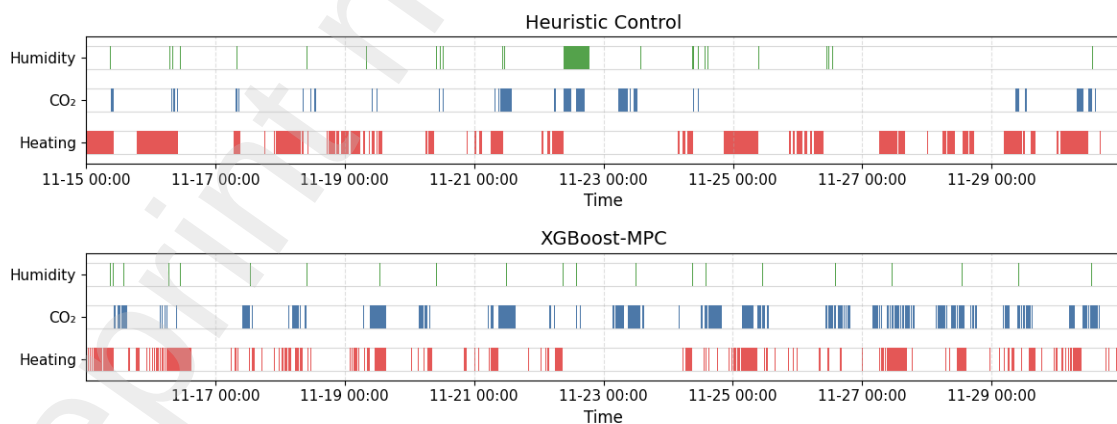


Fig. 10. Timelines of actuator switching over 16 days under XGBoost-MPC and heuristic control.

This difference translated into significantly improved environmental stability under MPC. The MPC achieved an RMSE of 1.67 °C and an MAE of 1.01 °C for indoor temperature, and an RMSE of 3.27% and an MAE of 2.65% for relative humidity. In contrast, the heuristic control showed substantially larger deviations (temperature: RMSE = 6.26 °C, MAE = 4.87 °C; humidity: RMSE = 14.66%, MAE = 11.92%). These results confirm that MPC improved microclimate regulation by a factor of three to five compared with the heuristic approach.

Building on this stability, plant-level regulation also improved, as illustrated in Fig. 11. The phenology-aware MPC, implemented with actuator constraints and stage-specific Q matrices, maintained growth length within a $\pm 10\%$ tolerance in 66.7% of evaluation cases, whereas the heuristic approach achieved only 28.6%. Together, these findings demonstrate that the MPC framework not only delivered finer environmental control but also translated this advantage into more reliable growth regulation.

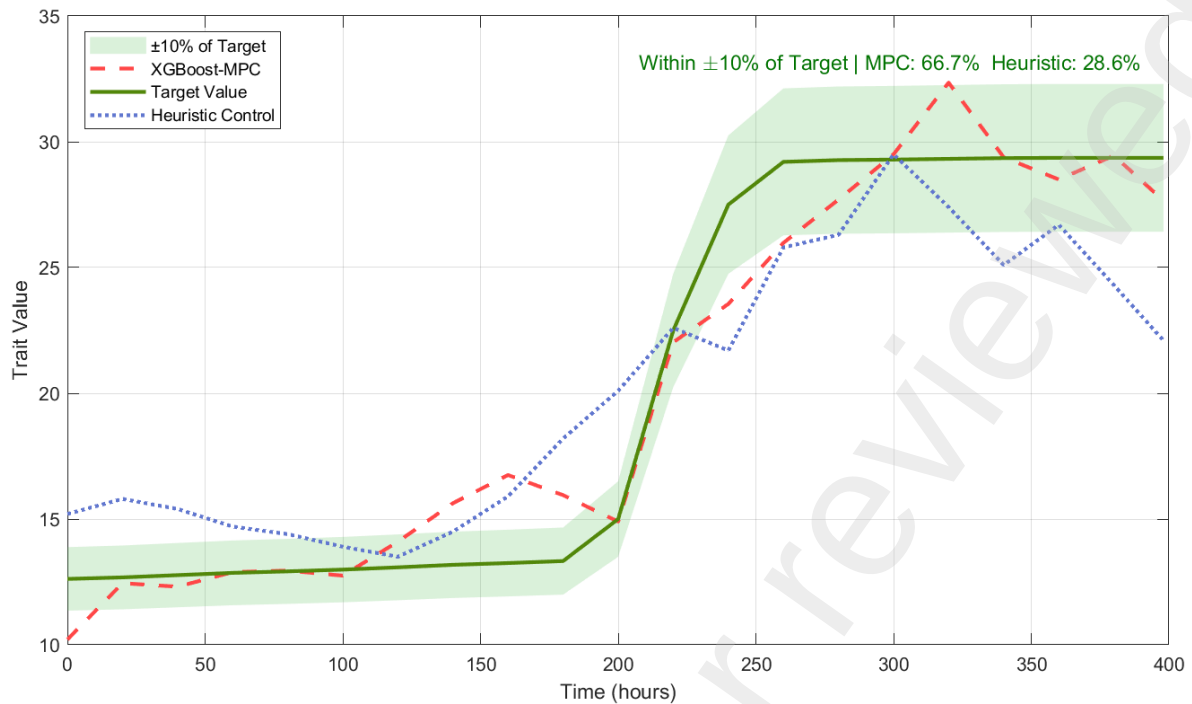


Fig. 11. Growth length control under XGBoost-MPC and heuristic strategies ($\pm 10\%$ band).

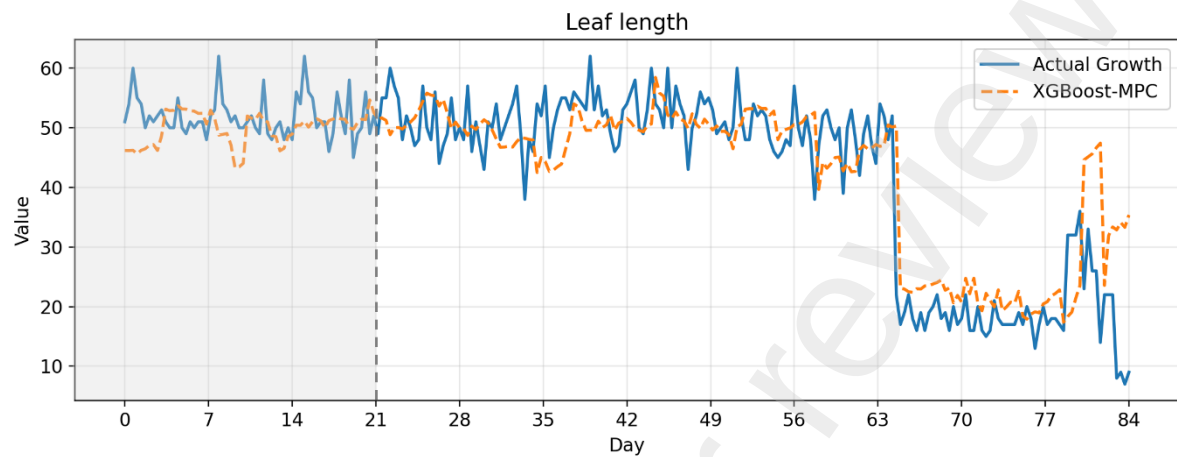
3.3.5 Model performance in on-site condition

To assess the long-term robustness of the GP, model forecasts were compared with empirical growth data collected over a 9-week cultivation period operating under predictor-based control. As illustrated in Fig. 12, the predicted and observed trajectories for representative morphological traits show a high degree of correlation, indicating that the model successfully captured the principal developmental patterns under dynamic, real-world cultivation conditions.

While minor deviations were observed at specific time points, the strong overall alignment confirms the model's stable forecasting performance throughout the extended cultivation period. Notably, this validation was conducted in a closed-loop system, meaning the observed data reflect the crop's actual response to the environment managed by the predictor. Therefore, these results substantiate both the forecasting accuracy of the model

and its efficacy in influencing realized growth outcomes within an autonomous control framework.

(a)



(b)

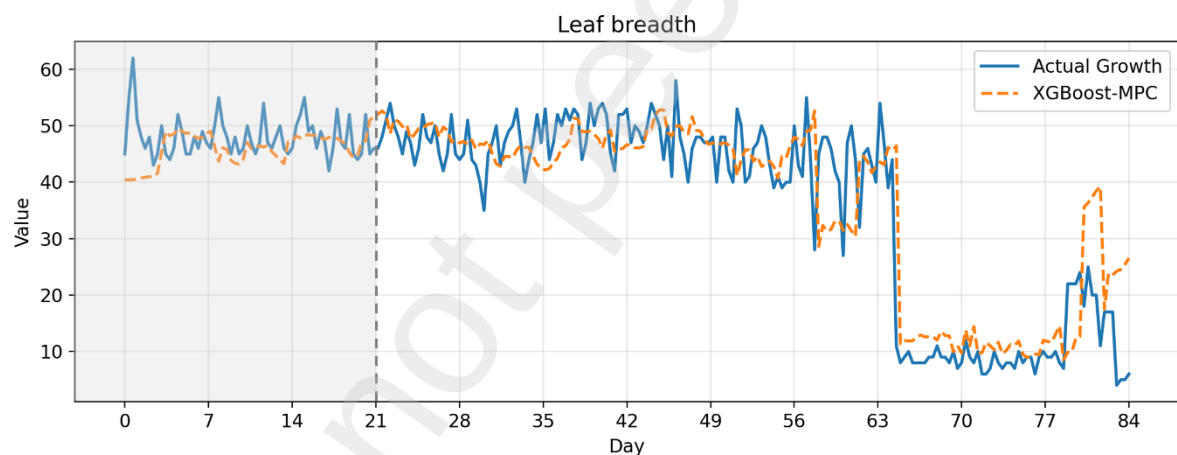


Fig. 12. Predicted vs. observed trajectories: (a) leaf length, (b) leaf breadth; shaded = weeks 1–3 validation.

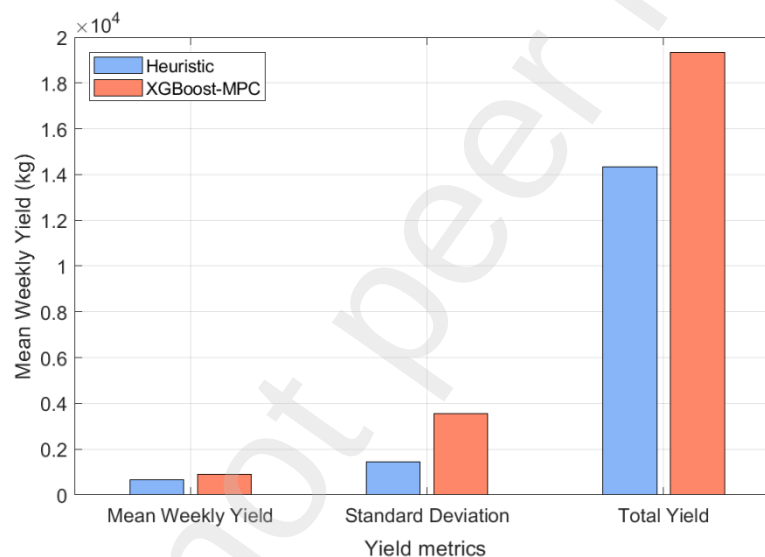
3.4 Yield and quality verification

An on-farm validation revealed significant productivity gains for the MPC strategy compared to a conventional heuristic approach. As detailed in Fig. 13 and Table 3, the MPC system achieved a total yield of 19,349 kg, a 35.1% increase over the 14,322 kg produced under heuristic control. Analysis of the weekly yield profiles also demonstrated that MPC

501 resulted in more consistent production rates with lower temporal variability.

502 Beyond total yield, the MPC strategy also substantially enhanced the marketable
 503 quality of the produce. The proportion of premium-grade fruits (defined as $\geq 2L$) rose from
 504 26.4% (3,787 fruits) under the heuristic method to 46.2% (8,944 fruits) with MPC. This
 505 represents a 2.36-fold increase in the absolute volume of premium fruits, underscoring the
 506 ability of the predictive control system to improve both crop productivity and commercial
 507 value.

(a)



(b)

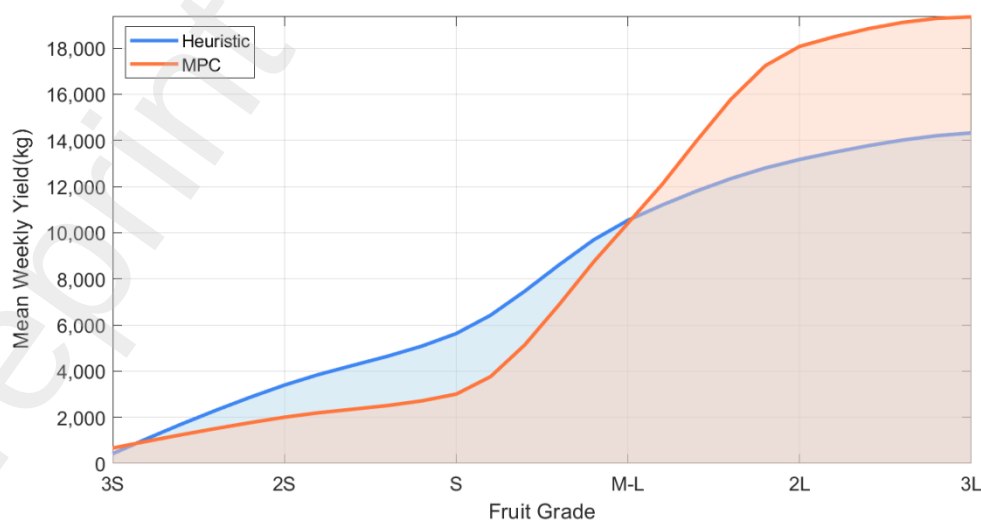


Fig. 13. Comparative yield performance of MPC and Heuristic: (a) key performance indicators; (b) weekly yield by fruit grade.

Table 3

Yield and quality comparison between heuristic and MPC control.

	Heuristic Control	MPC Control
Mean Weekly Yield	667 kg/week	906 kg/week
Total Yield	14,322 kg	19,349 kg
Quality Grade ($\geq 2L$)	26.4%	46.2%

4. Discussion

The proposed digital twin (DT)-based greenhouse control framework effectively replicated the physical greenhouse environment and demonstrated clear benefits in microclimate regulation, yield improvement, and fruit quality enhancement. By integrating the virtual-control module (VCM), environment twin (ET), and growth predictor (GP) in a closed loop, the framework provided significantly more stable control than heuristic strategies, highlighting the need for predictive, DT-driven methods in greenhouse management. Earlier studies (Priya et al., 2024; Elashmawy and Uysal, 2023; Yonbawi et al., 2023; Isied et al., 2022; Zhang et al., 2023; Hull et al., 2023; Liu et al., 2023) often addressed either prediction or simulation in isolation. In contrast, this study implemented a fully synchronized DT capable of real-time monitoring, prediction, and control. This provided novel insights into crop-environment interactions and allowed adaptive, stage-

specific regulation, unlike static threshold-based methods.

Data preprocessing and algorithm evaluation also proved critical. Ensemble tree-based methods, particularly XGBoost, achieved the most reliable forecasts compared to both traditional ensembles and deep learning models, showing the importance of tailored algorithm selection. The phenology-aware MPC successfully translated these predictions into actionable control without compromising crop physiology, ensuring practical stability.

However, several limitations remain. The experiments were restricted to tomato crops and a single cultivation period, limiting generalizability. Energy demand predictions could not be fully validated due to a lack of direct metering. Trait-specific variability, especially in reproductive features, suggests the need for multimodal data (e.g., image-based monitoring). Moreover, while MPC improved control precision, real-time synchronization across multiple greenhouses may pose scalability challenges.

Future work should expand validation across crops and seasons, incorporate multimodal sensing, and explore reinforcement learning-based control to further improve adaptability. Integrating energy management and deploying the framework in large-scale facilities will strengthen practical applicability. In practice, the DT-based framework can be embedded into farm management systems for adaptive, real-time decision-making. By synchronizing monitoring, prediction, and control, it provides a scalable methodology that enhances productivity, fruit quality, and sustainability in controlled-environment agriculture.

5. Conclusion

This study developed and validated an integrated digital twin framework for smart greenhouse management, comprising a VCM, an ET, and a data-driven GP. The VCM synchronized sensor data and actuator states with their virtual counterparts, while quantitative validation confirmed the ET's accuracy, with CV(RMSE) values for key

variables well below ASHRAE Guideline 14 thresholds. Correlation-based feature selection informed the design of a GP, within which XGBoost was identified as the most effective predictive engine. The MPC strategy enhanced environmental stability by reducing temperature and humidity deviations by 70–80% compared with heuristic control. On-farm verification further demonstrated a 35.1% increase in total yield and a more than twofold improvement in premium-grade fruit production, confirming the framework’s practical benefits for productivity and quality.

By integrating real-time monitoring, simulation, prediction, and actuation in a closed-loop cycle, the framework overcomes the limitations of heuristic strategies and provides a scalable methodology for precision agriculture. Future work should validate its applicability across diverse crops and seasons, incorporate multimodal sensing to improve trait-specific predictions, and explore reinforcement learning to enhance adaptability under dynamic conditions. Explicit integration of energy management and large-scale deployment will further strengthen its utility, enabling real-time decision support for sustainable, high-quality production in controlled-environment agriculture.

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References

- Alves, R.G., Maia, R.F., Lima, F., 2023. Development of a digital twin for smart farming: Irrigation management system for water saving. *J. Clean. Prod.* 388, 135920. <https://doi.org/10.1016/j.jclepro.2023.135920>.
- ASHRAE Guideline 14-2014: Measurement of Energy, Demand, and Water Savings. American Society of Heating, Refrigerating and Air-Conditioning Engineers.
- Asfarian, A., Wulandari, W., 2023. Digital twin platform architecture design to support smart aeroponic potato cultivation. *Proc. Bus. Innov. Eng. Conf. (BIEC)* 236, 92. https://doi.org/10.2991/978-94-6463-144-9_9.
- Assimakopoulos, F., Vassilakis, C., Margaris, D., Kotis, K., Spiliotopoulos, D., 2025. AI and related technologies in the fields of smart agriculture: a review. *Information* 16(2), 100. <https://doi.org/10.3390/info16020100>.
- Bissadu, K.D., Sonko, S., Hossain, G., 2025. Society 5.0 enabled agriculture: Drivers, enabling technologies, architectures, opportunities, and challenges. *Inf. Process. Agric.* 12(1), 112–124. <https://doi.org/10.1016/j.inpa.2024.04.003>.
- Breiman, L., 2001. Random forests. *Mach. Learn.* 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>.
- Çam, N.Y., Ezan, M.A., Biçer, Y., 2024. An integrated transient model in TRNSYS for thermal management of the tomato growth process in a greenhouse with a PV-aided heat pump-assisted HVAC system. *Int. J. Thermofluids* 22, 100592. <https://doi.org/10.1016/j.ijft.2024.100592>.
- Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system. *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.* 785–794. <https://doi.org/10.1145/2939672.2939785>.
- Choi, Y.B., Lee, I.B., 2025. Evaluation of evapotranspiration models integrating convolutional neural network-predicted leaf area for Pak Choi (*Brassica campestris* ssp. *chinensis*) in greenhouse environments. *Sci. Hortic.* 350, 114339. <https://doi.org/10.1016/j.scienta.2025.114339>.
- Devarajan, Y., 2025. Investigation of emerging technologies in agriculture: An in-depth look at smart farming, nano-agriculture, AI, and big data. *J. Biosyst. Eng.* 1–23. <https://doi.org/10.1007/s42853-025-00258-z>.
- Elashmawy, R., Uysal, I., 2023. Precision agriculture using soil sensor driven machine learning for smart strawberry production. *Sensors* 23(4), 2247. <https://doi.org/10.3390/s23042247>.
- Fan, X.R., Kang, M.Z., Heuvelink, E., De Reffye, P., Hu, B.G., 2015. A knowledge-and-data-driven modeling approach for simulating plant growth: A case study on tomato growth. *Ecol. Model.* 312, 363–373.

<https://doi.org/10.1016/j.ecolmodel.2015.06.006>.
 Foutia, A.A., Bouhounali, A., Bouhoun, S., Adnane, M., Benseddik, A., 2024. Development of an end-to-end solution for small farm monitoring and control. *Proc. Int. Conf. Embedded Distrib. Syst. (EDiS)* 333–337. <https://doi.org/10.1109/EDiS63605.2024.10783387>.
 Geurts, P., Ernst, D., Wehenkel, L., 2006. Extremely randomized trees. *Mach. Learn.* 63(1), 3–42. <https://doi.org/10.1007/s10994-006-6226-1>.
 Ghandar, A., Ahmed, A., Zulfiqar, S., Hua, Z., Hanai, M., Theodoropoulos, G., 2021. A decision support system for urban agriculture using digital twin: A case study with aquaponics. *IEEE Access* 9, 35691–35708.
 Guo, T., Peng, Y., Han, L., Jia, T., Zhang, C., Liu, W., Yang, Q., Huang, H., Hu, D., 2025. MAF-YOLOv8: A lightweight, high-precision deep learning model applied to real-time detection and counting of *Betula luminifera* seedling leaves. *Ind. Crops Prod.* 235, 121716. <https://doi.org/10.1016/j.indcrop.2025.121716>.
 Hamidane, H., Faiz, S.E., Rkik, I., El Khayat, M., Guerbaoui, M., Ed-Dahhak, A., Lachhab, A., 2024. Constrained temperature and relative humidity predictive control: Agricultural greenhouse case of study. *Inf. Process. Agric.* 11(3), 409–420. <https://doi.org/10.1016/j.inpa.2023.04.003>.
 Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. *Neural Comput.* 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>.
 Hosny, K.M., El-Hady, W.M., Samy, F.M., 2025. Technologies, protocols, and applications of Internet of Things in greenhouse farming: A survey of recent advances. *Inf. Process. Agric.* 12(1), 91–111. <https://doi.org/10.1016/j.inpa.2024.04.002>.
 Howard, D.A., Ma, Z., Jørgensen, B.N., 2022. A case study of digital twin for greenhouse horticulture production flow. *Proc. Int. Conf. Digital Twins Parallel Intell. (DTPI)* 1–6. <https://doi.org/10.1109/DTPI55838.2022.9998914>.
 Huang, W., Yang, H., Wang, Y., Ding, P., Nawi, N.M., Zhang, X., 2025. In-situ precision sensing for smart agriculture using multi-electrode sensor array systems in orchards. *Sens. Actuators A Phys.* 382, 116134. <https://doi.org/10.1016/j.sna.2024.116134>.
 Hull, K., Booysen, M.J., Mabitsela, M., Phiri, E., 2023. Using a digital twin for greenhouse tunnel temperature management and prediction. *Proc. IEEE AFRICON*, 1–6. <https://doi.org/10.1109/AFRICON55910.2023.10293411>.
 Hodson, T.O., 2022. Root mean square error (RMSE) or mean absolute error (MAE): When to use them or not.

- 632 Geosci. Model Dev. Discuss., 1–10. <https://doi.org/10.5194/gmd-15-5481-2022>.
- 633 Isied, R.S., Mengi, E., Zohdi, T.I., 2022. A digital-twin framework for genomic-based optimization of an
 634 agrophotovoltaic greenhouse system. *Proc. R. Soc. A* 478(2267), 20220414.
 635 <https://doi.org/10.1098/rspa.2022.0414>.
- 636 Juleang, P., Mitatha, S., Chaiyasoonthorn, S., 2024. A reliable protocol for wireless sensor networks to support
 637 large-scale smart farm systems. *Proc. Int. Conf. Inf. Technol. (InCIT)* 665–668.
 638 <https://doi.org/10.1109/InCIT63192.2024.10810616>.
- 639 Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Liu, T.Y., 2017. LightGBM: A highly efficient
 640 gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* 3146–3154.
 641 <https://doi.org/10.54646/bije.004>.
- 642 Lanje, S., Kawitkar, R., Mishra, S., 2022. A sensing approach for automated and real-time crop prediction in
 643 the scope of smart farming. *BOHR Int. J. Eng.* 1(1), 18–23. <https://doi.org/10.54646/bije.004>.
- 644 Li, H., Mao, Y., Shi, H., Fan, K., Sun, L., Zaman, S., Wang, Y., 2024. Establishment of deep learning model
 645 for the growth of tea cutting seedlings based on hyperspectral imaging technique. *Sci. Hortic.* 331, 113106.
 646 <https://doi.org/10.1016/j.scienta.2024.113106>.
- 647 Liaw, A., Wiener, M., 2002. Classification and regression by randomForest. *R News* 2(3), 18–22.
- 648 Liu, J., Wang, L., Wang, Y., Xu, S., Liu, Y., 2023. Research on the interface of sustainable plant factory based
 649 on digital twin. *Sustainability* 15(6), 5010. <https://doi.org/10.3390/su15065010>.
- 650 Manikandan, R., Ranganathan, G., Bindhu, V., 2022. Intelligent computing and control framework for smart
 651 automated system. *Intell. Autom. Soft Comput.* 33(1). <https://doi.org/10.32604/iasc.2022.023922>.
- 652 Miller, T., Mikiciuk, G., Durlik, I., Mikiciuk, M., Łobodzińska, A., Śnieg, M., 2025. The IoT and AI in
 653 agriculture: The time is now—A systematic review of smart sensing technologies. *Sensors* 25(12), 3583.
 654 <https://doi.org/10.3390/s25123583>.
- 655 Nimma, D., Dhanke, J.A., Murthy, G.S.N., Khandekar, S.D., Hymavathi, J., Jangir, P., Singh, M., 2025.
 656 Optimization of crop yields in sustainable agriculture: Application of big data analytics and artificial
 657 intelligence. *Remote Sens. Earth Syst. Sci.* 1–12. <https://doi.org/10.1007/s41976-024-00189-x>.
- 658 Pal, S., VijayKumar, H., Akila, D., Jhanjhi, N.Z., Darwish, O.A., Amsaad, F., 2023. Information-centric IoT-
 659 based smart farming with dynamic data optimization. *Comput. Mater. Continua* 74(1), 321–338.
 660 <https://doi.org/10.32604/cmc.2023.029038>.
- 661 Priya, S.S., Akilesh, R., Karthikeyan, S., Balasabarish, M., GD, S., 2024. IoT-based precision agriculture using

smart sensors and cloud computing for crop monitoring and yield optimization. *Proc. Int. Conf. Innov. Comput. Intell. Commun. Smart Electr. Syst. (ICSES)* 1–7. <https://doi.org/10.1109/ICSES63760.2024.10910911>.

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., Gulin, A., 2018. CatBoost: Unbiased boosting with categorical features. *Adv. Neural Inf. Process. Syst.*, 6638–6648.

Rao, G.B.N., Rao, K.V., Kamarajugadda, R., Reddy, A.A., Rani, P.P., 2023. Smart farming for agriculture management using IoT. *Proc. Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)* 1, 540–544. <https://doi.org/10.1109/ICACCS57279.2023.10112839>.

Rizan, N., Balasundram, S.K., Shahbazi, A.B., Balachandran, U., Shamshiri, R.R., 2024. Internet-of-Things for smart agriculture: Current applications, future perspectives, and limitations. *Agric. Sci.* 15(12), 1446–1475. <https://doi.org/10.4236/as.2024.1512080>.

Rushchitskaya, O., Kulikova, E., Kot, E., Kruzhkova, T., 2024. Sustainable practices and technological innovations transforming agribusiness dynamics. *E3S Web Conf.* 542, 03003. <https://doi.org/10.1051/e3sconf/202454203003>.

Sahu, S.K., Liu, H., 2023. A genetic solution for the global food security crisis. *J. Integr. Plant Biol.* 65(6), 1359–1361. <https://doi.org/10.1111/jipb.13500>.

Salimath, M., Kaliannan, N., Prabhakar, V., Iyyakutty, R., Jeyabaskaran, K.J., 2025. IoT and sensor technologies: Increased water and nutrient savings and profit in Banana cv. Grand Nain (AAA) production. *Sci. Hortic.* 341, 113982. <https://doi.org/10.1016/j.scienta.2025.113982>.

Sari, B.G., Lúcio, A.D.C., Santana, C.S., Savian, T.V., 2019. Describing tomato plant production using growth models. *Sci. Hortic.* 246, 146–154. <https://doi.org/10.1016/j.scienta.2018.10.044>.

Schlagmann, M., Stoenner, J., Selbmann, F., Hess, S., Otto, T., 2023. Greenhouse monitoring with biocompatible humidity sensor for smart farming. *Proc. Smart Syst. Integration Conf. Exhib. (SSI)* 1–6. <https://doi.org/10.1109/SSI58917.2023.10387759>.

Sharma, N., Rout, S., Saritha, B., Gautam, S.K., 2024. A comprehensive review on future of smart farming and its role in shaping food production. *J. Exp. Agric. Int.* 46(5), 486–493. <https://doi.org/10.9734/JEAI/2024/v46i52401>.

Singh, S.P., Dhiman, G., Juneja, S., Viriyasitavat, W., Singal, G., Kumar, N., Johri, P., 2023. A new QoS optimization in IoT-smart agriculture using rapid-adaption-based nature-inspired approach. *IEEE Internet Things J.* 11(3), 5417–5426. <https://doi.org/10.1109/JIOT.2023.3306353>.

692 Spyrou, O., Verdouw, C., Hurst, W., 2024. A digital twin reference architecture for pharmaceutical cannabis
693 production. *Int. J. Comput. Integr. Manuf.* 37(6), 726–746.
694 <https://doi.org/10.1080/0951192X.2023.2257635>.

695 Srisawat, T., Sakprom, S., Kunsawat, P., Praksong, K., Suchat, S., Muangprathub, J., 2025. IoT-enabled
696 agricultural environmental monitoring: Enhancing growth and yield using natural-rubber straw and
697 mulching experiment. *Ind. Crops Prod.* 225, 120524. <https://doi.org/10.1016/j.indcrop.2025.120524>.

698 Unity Technologies, 2023. Unity Manual. Unity Technologies.
699 https://docs.unity3d.com/Manual/index.html?utm_source=chatgpt.com (accessed 13 August 2025)

700 U.S. Department of Energy, 2024. EnergyPlus™ Engineering Reference — The Reference to EnergyPlus
701 Calculations, Version 24.2. U.S. Department of Energy. [https://bigladdersoftware.com/epx/docs/24-](https://bigladdersoftware.com/epx/docs/24-2/engineering-reference/?utm_source=chatgpt.com)
702 [2/engineering-reference/?utm_source=chatgpt.com](https://bigladdersoftware.com/epx/docs/24-2/engineering-reference/?utm_source=chatgpt.com) (accessed 13 August 2025)

703 Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017.
704 Attention is all you need. *Adv. Neural Inf. Process. Syst.* 5998–6008.

705 Vijai, C., Worakamol, W., Elayaraja, M., 2023. Climate change and its impact on agriculture. *Int. J. Agric. Sci.*
706 *Vet. Med.* 11(4). <https://doi.org/10.25303/1104ijasvm0108>.

707 Willmott, C.J., Matsuura, K., 2005. Advantages of the mean absolute error (MAE) over the root mean square
708 error (RMSE) in assessing average model performance. *Clim. Res.* 30(1), 79–82.
709 <https://doi.org/10.3354/cr030079>.

710 Yonbawi, S., Alahmari, S., Raju, B.R.S.S., Rao, C.H.G., Ishak, M.K., Alkahtani, H.K., Mostafa, S.M., 2023.
711 Modeling of sensor enabled irrigation management for intelligent agriculture using hybrid deep belief
712 network. *Comput. Syst. Sci. Eng.* 46(2). <http://doi.org/10.32604/csse.2023.036721>.

713 Yassine, H.M., Roufaida, K., Shkodyrev, V.P., Abdelhak, M., Zarour, L., Khaled, R., 2022. Intelligent farm
714 based on deep reinforcement learning for optimal control. *Proc. Int. Symp. Innovative Informatics Biskra*
715 *(ISNIB)* 1–6. <https://doi.org/10.1109/ISNIB57382.2022.10076088>.

716 Zhang, Z., Zhu, Z., Gao, G., Qu, D., Zhong, J., Jia, D., Pan, S., 2023. Design and research of digital twin system
717 for multi-environmental variable mapping in plant factory. *Comput. Electron. Agric.* 213, 108243.
718 <https://doi.org/10.1016/j.compag.2023.108243>.